DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III Topics in Analysis (3TN)

Revision on sequences of real or complex numbers

Dr Francois Ebobisse

22 July 2024

Definition 0.1 (Sequence)
Let X be a non-empty set. A sequence of elements in X is a function a function $x:\mathbb{N}\to X$. The elements $x(0)$, $x(1)$, $x(2),\cdots$ of the sequence are denoted by x_0, x_1, $x_2,\cdots$ and the sequence $x:\mathbb{N}\to X$ is often denoted by $(x_n)_{n\in\mathbb{N}}$ or simply (x_n).
We say that (y_k) is a subsequence of (x_n) if there exists a strictly increasing sequence of positive integers (n_k) such that $y_k = x_{n_k}$ for any $k\in\mathbb{N}$.

Remark 0.2

(i) In this course, we will study sequences taking values in the set $\mathbb{R}$ of real numbers or in the set $\mathbb{C}$ of complex numbers.

(ii) Instead of the whole set of positive integers $\mathbb{N}$, we will often study sequences defined for only positive integers greater than a certain fixed integers, that is $x:\{n\in\mathbb{N}\colon\ n\geq n_0\}\to X$.

Example 0.3

(i) $\left(\dfrac{1}{n}\right)$.

(ii) Let $z\in\mathbb{C}$ and consider the sequence (z^n). Enumerate the elements of the sequence when $z=1,\,-1,\,i$

(iii) Let $z\in\mathbb{C}$ and $x_n := \displaystyle\sum_{k=0}^{n} z^k$. It is clear that

$$\boxed{x_n = \begin{cases} \dfrac{1-z^{n+1}}{1-z} & \text{if } z\neq 1, \\[2ex] n+1 & \text{if } z = 1\,. \end{cases}}$$

(iv) A sequence can also be described inductively. For instance,

$$\boxed{\begin{cases} x_0 = 1, \\[2ex] x_{n+1} = 1 + \dfrac{1}{x_n}\,. \end{cases}}$$

Notice that in this case, it is not easy to find the sequence as a function of n only.

Definition 0.4 (Bounded sequence)

(i) We say that the sequence $(x_n) \subset \mathbb{R}$ is bounded from below if the set $(x_0, x_1, \cdots, \}$ is bounded from below in $\mathbb{R}$. That is, there exists a real number K such that $x_n \geq K$ for any $n \in \mathbb{N}$.

(ii) We say that the sequence $(x_n) \subset \mathbb{R}$ is bounded from above if the set $(x_0, x_1, \cdots,)$ is bounded from above in $\mathbb{R}$. That is, there exists a real number M such that $x_n \leq M$ for any $n \in \mathbb{N}$.

(iii) We say that the sequence $(x_n) \subset \mathbb{C}$ is bounded if the sequence $(|x_n|)$ is bounded from above. That is, there exists a real number $M > 0$ such that $|x_n| \leq M$ for any $n \in \mathbb{N}$.

We are more interested in the behaviour of a sequence for large values of the indices n. Therefore we introduce the notion of limit of a sequence.

Definition 0.5 (Limit of a sequence)
Let $(x_n) \subset \mathbb{C}$ and let $\ell \in \mathbb{C}$. We say that ℓ is the limit of the sequence (a_n) when n goes to ∞ or that the sequence (a_n) converges to ℓ when n goes to ∞ if the following holds:

$$\boxed{\forall \varepsilon > 0\ \exists n_\varepsilon \in \mathbb{N} : \quad n \geq n_\varepsilon \quad \Rightarrow \quad |x_n - \ell| < \varepsilon}\,.$$

We write $\lim\limits_{n\to\infty} x_n = \ell$ or $x_n \to \ell$ for $n \to \infty$.

Remark 0.6

(i) If the sequence (x_n) is real then $\ell \in \mathbb{R}$ and the definition of the limit does not change. In that case the modulus $|x_n - \ell|$ is the absolute value.

(ii) In the case of real sequences, there are also the notions of liminf and limsup and of sequences diverging to $+\infty$ or $-\infty$.

(1) We say that $\ell' = \liminf\limits_{n\to\infty} x_n$ if

$$\boxed{\ell' = \sup_{k\in\mathbb{N}} \inf_{n\geq k} x_n \ \Leftrightarrow\ \forall k \in \mathbb{N}\ \inf_{n\geq k} x_n \leq \ell' \text{ and } \forall \varepsilon > 0\ \exists k_\varepsilon \in \mathbb{N} :\ n \geq k_\varepsilon \ \Rightarrow\ \ell' - \varepsilon < x_n}\,.$$

(2) We say that $\ell'' = \limsup\limits_{n\to\infty} x_n$ if

$$\boxed{\ell'' = \inf_{k\in\mathbb{N}} \sup_{n\geq k} x_n \ \Leftrightarrow\ \forall k \in \mathbb{N}\ \sup_{n\geq k} x_n \geq \ell'' \text{ and } \forall \varepsilon > 0\ \exists k_\varepsilon \in \mathbb{N} :\ n \geq k_\varepsilon \ \Rightarrow\ \ell'' + \varepsilon > x_n}\,.$$

(3) It is easy to see that $\ell' \leq \ell''$.

(4) One can see that the sequence (x_n) converges to ℓ when $n \to \infty$ if $\ell' = \ell'' = \ell$.

(5) We say that the sequence (x_n) has limit $+\infty$ when $n \to \infty$ or that the sequence (x_n) diverges positively when n goes to $+\infty$, if

$$\boxed{\forall K > 0\ \exists n_K \in \mathbb{N}:\ n \geq n_K \quad \Rightarrow \quad x_n > K\,.}$$

Analogously, we say that the sequence (x_n) has limit $-\infty$ when $n \to \infty$ or that the sequence (x_n) diverges negatively when $n \to \infty$, if

$$\boxed{\forall K > 0\ \exists n_K \in \mathbb{N}:\ n \geq n_K \quad \Rightarrow \quad x_n < -K\,.}$$

Exercise 0.7

1. Show that if the sequence $(x_n) \subset \mathbb{C}$ converges to ℓ when $n \to \infty$, then any subsequence (y_k) of (x_n) converges ℓ.
2. Let (x_n) be a sequence of real numbers. Show if $\ell' = \liminf_{n\to\infty} x_n$, then there exists a subsequence (y_k) of (x_n) such that $\ell' = \lim_{k\to\infty} y_k$. Similarly for $\ell'' = \limsup_{n\to\infty} x_n$.
3. Show that if $(x_n) \subset \mathbb{C}$ converges to ℓ, then (x_n) is bounded. Is the converse true? Explain your answer.
4. Show that the limit of a converging sequence is unique.
5. Show that if (x_n) converges to $\ell \neq 0$, then there exists $C > 0$ such that $|x_n| \geq C$ for n large enough.

Theorem 0.8 (Algebraic properties of limits)
Let (x_n) and (y_n) be sequences of real or complex numbers. If $x_n \to \ell_1$ and $y_n \to \ell_2$ with ℓ_1 and ℓ_2 finite, then:

(i) $x_n + y_n \to \ell_1 + \ell_2$ *when* $n \to \infty$*;*

(ii) $x_n y_n \to \ell_1 \ell_2$ *when* $n \to \infty$*;*

(iii) $\dfrac{1}{y_n} \to \dfrac{1}{\ell_2}$ *when* $n \to \infty$ *provided* $\ell_2 \neq 0$*;*

(iv) $\dfrac{x_n}{y_n} \to \dfrac{\ell_1}{\ell_2}$ *when* $n \to \infty$ *provided* $\ell_2 \neq 0$*;*

Proof: Let $\varepsilon > 0$. There exist n_ε^1, $n_\varepsilon^2 \in \mathbb{N}$ such that

$$n \geq n_\varepsilon^1 \;\Rightarrow\; |x_n - \ell_1| < \frac{\varepsilon}{2} \quad \text{and} \quad n \geq n_\varepsilon^2 \;\Rightarrow\; |y_n - \ell_2| < \frac{\varepsilon}{2}$$

Let $n_\varepsilon = n_\varepsilon^1 + n_\varepsilon^2$. Then $n \geq n_\varepsilon \;\Rightarrow\; |x_n + y_n - (\ell_1 + \ell_2)| \leq |x_n - \ell_1| + |y_n - \ell_2| < \dfrac{\varepsilon}{2} + \dfrac{\varepsilon}{2} = \varepsilon$ and this proves (i).
For $n \geq n_\varepsilon$, we have

$$|x_n y_n - \ell_1 \ell_2| = |x_n y_n - \ell_1 y_n + \ell_1 y_n - \ell_1 \ell_2| \leq |x_n - \ell_1| \cdot |y_n| + |\ell_1| \cdot |y_n - \ell_2| < \frac{\varepsilon}{2}(|y_n| + |\ell_1|)$$

Now from Exercise 0.7 (3), there exists $M \geq 0$ such that $|y_n| \leq M$ for any n. Therefore we get $n \geq n_\varepsilon \Rightarrow |x_n y_n - \ell_1 \ell_2| < \frac{\varepsilon}{2}(M + |\ell_1|)$ which proves (ii).
To prove (iii), we use Exercise 0.7 (5) and get $n_\varepsilon^3 \in \mathbb{N}$ and a constant $C > 0$ such that $|y_n| \geq C$ for any $n \geq n_\varepsilon^3$. Now setting $n_\varepsilon = n_\varepsilon^1 + n_\varepsilon^2 + n_\varepsilon^3$ we get

$$n \geq n_\varepsilon \quad \Rightarrow \quad \left| \frac{1}{y_n} - \frac{1}{\ell_2} \right| = \frac{|\ell_2 - y_n|}{|\ell_2| \cdot |y_n|} < \frac{\varepsilon}{2C|\ell_2|},$$

$$\begin{aligned} n \geq n_\varepsilon \quad \Rightarrow \quad \left| \frac{x_n}{y_n} - \frac{\ell_1}{\ell_2} \right| &= \frac{|\ell_2 x_n - \ell_1 y_n|}{|\ell_2| \cdot |y_n|} = \frac{|\ell_2 x_n - \ell_1 \ell_2 + \ell_1 \ell_2 - \ell_1 y_n|}{|\ell_2| \cdot |y_n|} \\ &\leq \frac{|\ell_2| \cdot |x_n - \ell_1| + |\ell_1| \cdot |\ell_2 - y_n|}{|\ell_2| \cdot |y_n|} < \frac{\varepsilon}{2C|\ell_2|}(|\ell_1| + |\ell_2|) \end{aligned}$$

which prove (iii) and (iv). $\square$

Theorem 0.9 (Comparison)
Let (x_n) and (y_n) be two sequences of real numbers. If $x_n \to \ell_1$ and $y_n \to \ell_2$ for $n \to \infty$ and if $x_n \le y_n$ for n large, then $\ell_1 \le \ell_2$.

Proof: Assume that ℓ_1 and ℓ_2 are finite and $\ell_1 > \ell_2$. Choosing $0 < \varepsilon < \frac{1}{2}(\ell_1 - \ell_2)$, then we have by assumption some n_ϵ **(explain why!)** such that

$$n \ge n_\varepsilon \quad \Rightarrow \quad x_n \le y_n,\ |x_n - \ell_1| < \varepsilon \text{ and }, |y_n - \ell_2| < \varepsilon\,.$$

Therefore for such n, we get $\ell_1 - \varepsilon < x_n \le y_n < \ell_2 + \varepsilon$ which implies that $0 < \ell_1 - \ell_2 < 2\varepsilon$ and contradicts the choice $0 < \varepsilon < \frac{1}{2}(\ell_1 - \ell_2)$.
The case $\ell_1 = \pm\infty$ or $\ell_2 = \pm\infty$ is obtained similarly. □

Exercise 0.10

1. Prove that if $x_n \to \ell$, then $|x_n| \to |\ell|$. Is the converse true? Explain your answer.
2. Prove that if $x_n \to 0$ and (y_n) is bounded, then $x_n \cdot y_n \to 0$.
3. Prove that if $x_n \to \ell \in \mathbb{C}$ or $\mathbb{R}$, then $\lim\limits_{n\to\infty}(x_{n+1} - x_n) = 0$. Is the converse true? Explain your answer.
4. Prove that if $x_n \to \ell \neq 0$ with $\ell \in \mathbb{C}$ or $\mathbb{R}$, then $\lim\limits_{n\to\infty} \dfrac{x_{n+1}}{x_n} = 1$. Is the converse true? Explain your answer. What happens when $\ell = 0,\ \pm\infty$?
5. Let (x_n) and (y_n) be sequences of real numbers. Prove that
   (a) If $x_n \to \infty$ and (y_n) is bounded from below, then $x_n + y_n \to \infty$;
   (b) If $x_n \to -\infty$ and (y_n) is bounded from above, then $x_n + y_n \to -\infty$.
6. Let (x_n), (y_n) and (z_n) be sequences of real numbers such that $x_n \le y_n \le z_n$ for n large. Prove that if $x_n \to \ell$ and $z_n \to \ell$ (with $\ell \in \mathbb{R}$ or $\ell = \pm\infty$), then $y_n \to \ell$.

Definition 0.11 (Increasing/decreasing sequence)
Let (x_n) be a sequence or real numbers. We say that (x_n) is increasing if $x_{n+1} \ge x_n$ for any $n \in \mathbb{N}$. We say that (x_n) is decreasing if $x_{n+1} \le x_n$ for any $n \in \mathbb{N}$. We say that (x_n) is strictly increasing or strictly decreasing if the properties above hold with strict.

Proposition 0.12 *Let $\{a_n\} \subset \mathbb{R}$ be a monotone sequence. Then*

$$\lim_{n\to\infty} a_n = \begin{cases} \sup\limits_{n\in\mathbb{N}} a_n \in]-\infty, +\infty] & \textit{if } \{a_n\} \textit{ is increasing,} \\ \\ \inf\limits_{n\in\mathbb{N}} a_n \in [-\infty, +\infty[& \textit{if } \{a_n\} \textit{ is decreasing.} \end{cases}$$

In particular, a monotone sequence is convergent if and only if is bounded.

Proof:(Exercise!)

The study of a sequence of real numbers

We are going to study the behaviour of a sequence of real numbers (x_n) according to the following steps:

(i) We will check whether the given sequence (x_n) is well-defined.

(ii) We will check whether the sequence (x_n) is monotone.

(iii) We will check the boundedness of the sequence (x_n).

(iv) We will determine whether the sequence (x_n) converges or diverges and possibly find its limit $\ell \in \bar{\mathbb{R}}$.

Recursive sequences:

We will now focus on the study of recursive sequences in particular. Precisely, we are interested in sequences (x_n) defined as

$$\begin{cases} x_0 = a, \\ x_{n+1} = f(x_n) \quad \forall n \in \mathbb{N} \end{cases}$$

where a is a given real number and f is a given function. So, we get that

$$x_1 = f(a), \quad x_2 = f(x_1) = f(f(a)), \; \ldots \; .$$

Therefore, the sequence (x_n) is obtained by iteration of the function f startimg from the number a. In general, we will not get x_n explicitely as a funtion of n and this is not a handicap for the study of the sequence (x_n).

Example 0.13

- **Arithmetic sequence:** a sequence of numbers in which the difference between consecutive terms is always the same. That is, (a_n) defined by $a_{n+1} = a_n + d$ for every $n \in \mathbb{N}$.

- **Geometric sequence:** a sequence of non-zero numbers where each term after the first is found by multiplying the previous one by a fixed, non-zero number called the common ratio. That is, (g_n) is such that $g_0 = a$ and $g_{n+1} = r g_n$ with $r \neq 0$ for every $n \in \mathbb{N}$.

Notice that for both cases, the sequence can be expressed as a function of n. In fact,

$$\boxed{a_n = a_0 + nd \quad \text{and} \quad g_n = ar^n}.$$

Notice that, in general recursive sequences are not always expressed explicitely as functions of n.

Remark 0.14 Notice that a recursive sequence is not always defined. In fact, consider

$$\begin{cases} x_0 = 1, \\ x_{n+1} = 1 - \frac{1}{x_n} \quad \forall n \in \mathbb{N} \end{cases}$$

and notice that $x_1 = 0$ and x_2 cannot be defined!

The following proposition provides conditions under which a sequence si defined.

Proposition 0.15 *Let $f : E \subset \mathbb{R} \to \mathbb{R}$ be any function and let $a \in E$. If $f(E) \subset E$, then the sequence (x_n) defined by*

$$\begin{cases} x_0 = a, \\ x_{n+1} = f(x_n) \quad \forall n \in \mathbb{N} \end{cases}$$

is well-defined and $(x_n) \subset E$.

Clearly, in order to apply this proposition, one just needs to make a suitable choice of the subset E.

(a) The sequence (x_n) defined by

$$\begin{cases} x_0 = 0, \\ x_{n+1} = e^{x_n} \quad \forall n \in \mathbb{N} \end{cases}$$

is well-defined since the function $f(x) = e^x$ is defined everywhere and $f(\mathbb{R}) \subset \mathbb{R}$.

(b) The sequence (x_n) defined by

$$\begin{cases} x_0 = 2, \\ x_{n+1} = \ln(x_n) + 1 \quad \forall n \in \mathbb{N} \end{cases}$$

is well-defined. In fact, for $E := [1, \infty)$, function $f(x) = \ln(x) + 1$ is such that $f(E) \subset E$, since

$$x \geq 1 \quad \Rightarrow \quad \ln(x) \geq \ln(1) = 0 \quad \Rightarrow \quad f(x) = \ln(x) + 1 \geq 0 + 1 = 1\,.$$

Give a graphical description of a recursive sequence!

In the next proposition and corollary, we present the monotonicity of a recursive sequence.

Proposition 0.16 (Case where f is increasing)
Consider the recursive sequence (x_n) defined by

$$\begin{cases} x_0 = a, \\ x_{n+1} = f(x_n) \quad \forall n \in \mathbb{N}\,. \end{cases}$$

Assume that the function f is increasing. Then, the following facts hold.

- *If $x_0 \leq x_1$, then the sequence (x_n) is increasing.*
- *If $x_0 \geq x_1$, then the sequence (x_n) is decreasing.*

Proof: Assume that $x_0 \leq x_1$ and let us show that $x_n \leq x_{n+1}$ for every $n \in \mathbb{N}$. By induction, for $n = 0$ the statement is true by assumption. Assume that the statement is true for n and let us show that $x_{n+1} \leq x_{n+2}$. In fact

$$x_n \leq x_{n+1} \quad \Rightarrow \quad x_{n+1} = f(x_n) \leq f(x_{n+1}) = x_{n+2}\,.$$

A similary argument is applied to the case $x_0 \geq x_1$. ∎

Corollary 0.17 (Case where f is decreasing) *Consider the recursive sequence (x_n) defined by*

$$\begin{cases} x_0 = a, \\ x_{n+1} = f(x_n) \quad \forall n \in \mathbb{N}\,. \end{cases}$$

Assume that the function f is decreasing. Then, the following facts hold.

- *If $x_0 \le x_2$, then the sequence (x_{2n}) is increasing and (x_{2n+1}) is decreasing.*
- *If $x_0 \ge x_2$, then the sequence (x_{2n}) is decreasing and (x_{2n+1}) is increasing.*

Proof: First notice that setting $y_n = x_{2n}$ and $z_n = x_{2n+1}$, we get

$$y_{n+1} = x_{2n+2} = (f\circ f)(x_{2n}) = (f\circ f)(y_n) \quad \text{and} \quad z_{n+1} = x_{2n+3} = (f\circ f)(x_{2n+1}) = (f\circ f)(z_n)\,.$$

Hence, the two sequences (y_n) and (z_n) are recursive sequences defined by the function $f^2 = f\circ f$ which is an increasing function, since the function f is decreasing. Therefore, we can apply Proposition 0.16 to the sequences (y_n) and (z_n) and get the conclusion of the corollary. This completes the proof of the corollary. ■

Example 0.18 Consider the sequence (x_n) defined by

$$\boxed{\begin{cases} x_0 = 1, \\ x_n = \sqrt{x_{n-1}+6}, \quad n = 1, 2, 3, \ldots. \end{cases}}$$

Study the behaviour of the sequence (x_n).

Solution:

(i) Notice that $x_n > 0$ for every $n \in \mathbb{N}$ (you can show it by induction). Hence, $x_n \ge -6$.

(ii) Monotinicity: We can apply here Proposition 0.16, since the function $(x) = \sqrt{x+6}$ is increasing and that $x_1 = \sqrt{x_0+6} = \sqrt{7} > 1 = x_0$. Therefore, we can deduce that the sequence (x_n) is increasing.

We can also study the monotonicity of the sequence (x_n) directly. To this aim, we study the sign of $x_{n+1} - x_n$ for every $n \in \mathbb{N}$. Notice that

$$x_{n+1} - x_n = \sqrt{x_n+6} - \sqrt{x_{n-1}+6} = \boxed{\frac{[\sqrt{x_n+6} - \sqrt{x_{n-1}+6}][\sqrt{x_n+6} + \sqrt{x_{n-1}+6}]}{\sqrt{x_n+6} + \sqrt{x_{n-1}+6}}}$$

$$= \boxed{\frac{x_n - x_{n-1}}{\underbrace{\sqrt{x_n+6} + \sqrt{x_{n-1}+6}}_{=:D_n}}} \qquad (x_{n+1} - x_n \text{ has the same sign as } x_n - x_{n-1} \text{ since } D_n > 0)$$

$$= \frac{x_{n-1} - x_{n-2}}{D_n D_{n-1}} = \ldots = \frac{x_1 - x_0}{D_n D_{n-1} \ldots D_1} = \ldots = \frac{\sqrt{x_0+6} - x_0}{D_n D_{n-1} \ldots D_1} = \frac{\sqrt{7} - 1}{D_n D_{n-1} \ldots D_1} > 0\,.$$

Hence, $x_{n+1} - x_n > 0$ for every $n \in \mathbb{N}$ and therefore, the sequence (x_n) is strictly increasing.

(iii) Boundedness: being the sequence increasing either $\lim_{n\to\infty} x_n = \infty$ or the sequence (x_n) is bounded and hence, converges to a number $\ell > 0$. The number ℓ is the natural upper bound of the sequence (x_n), that is, $x_n \leq \ell$ for every $n \in \mathbb{N}$. In this case, if the sequence (x_n) converges to ℓ, then we get

$$x_n = \sqrt{x_{n-1}+6} \quad \Rightarrow \quad \ell = \sqrt{\ell+6} \quad \Rightarrow \quad \ell = 3\,.$$

So, let us check whether $x_n \leq 3$ for every $n \in \mathbb{N}$. By induction, for $n = 0$, we get that $x_0 = 1 < 3$. Now assume that $x_n \leq 3$ and let us show that $x_{n+1} \leq 3$.

$$x_{n+1} = \sqrt{x_n+6} \leq \sqrt{3+6} = \sqrt{9} = 3\,.$$

Notice that this is not the only way of finding an upper bound to the sequence (x_n). One could also guess an upper bound M and then prove by induction (as above) that the number M is indeed an upper bound to the sequence.

(iv) We have been able to establish that the sequence (x_n) is increasing and is bounded from above. Hence, (x_n) converges to $\ell = 3$ (as obtained in (iii)).

Newton method: Besides the bisection method which is a root-finding method that applies to any continuous functions for which one knows two values with opposite signs, there is another well-known method called the Newton method which generates a recursive sequence. Precisely, let $f : [a, b] \to \mathbb{R}$ be a continuously differentiable function and let $x_0 \in (a, b]$. The recursive sequence generated through the Newton method is described as follows: Assume that we know x_n. Then, the following term x_{n+1} is obtained by considering the intersection with the x-axis of the tangent line to the graph of the function f at the number $x = x_n$. We recall that the cartesian equation of the tangent line to the graph of the function f at the number $x = x_n$ is

$$y = f(x_n) + f'(x_n)(x - x_n)\,.$$

We set $y = 0$ and solve x (assuming that $f'(x_n) \neq 0$) and get $x = x_n - \dfrac{f(x_n)}{f'(x_n)}$. Now, we set

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}}\,.$$

Exercises 0.19

1. Study the behaviour of the sequence (x_n) defined by

$$\boxed{\begin{cases} x_0 = 1, \\ x_n = \sqrt{2x_{n-1}+3}, \quad n = 1, 2, 3, \ldots. \end{cases}}\,.$$

2. Consider the sequence (a_n) defined by

$$\boxed{\begin{cases} a_0 = 1, \\ a_{n+1} = \dfrac{1}{2}\left(a_n + \dfrac{2}{a_n}\right), \quad n = 0, 1, 2, 3, \ldots. \end{cases}}\,.$$

(a) Show that $a_n \geq \sqrt{2}$ for every $n = 1, 2, 3, \ldots$.

(b) Show that $a_n \geq a_{n+1}$.

(c) Deduce that the sequence (a_n) converges to $\ell = \sqrt{2}$.

3. What can you say about the behaviour of each of the sequences (u_n) and (v_n) defined by

$$\boxed{\begin{cases} u_0 = \alpha > 0, \\ u_{n+1} = u_n + \dfrac{1}{u_n}, \quad n = 0, 1, 2, 3, \ldots. \end{cases}}$$

and

$$\boxed{\begin{cases} v_0 = \beta \in [0, \frac{\pi}{2}], \\ v_{n+1} = v_n \sin(v_n), \quad n = 0, 1, 2, 3, \ldots. \end{cases}}\, ?$$